

Born approximation 3

We will now show why we can use the approximation

$$\frac{\exp(ik|\mathbf{x} - \mathbf{y}|)}{|\mathbf{x} - \mathbf{y}|} V(\mathbf{y}) \approx \frac{1}{|\mathbf{x}|} \exp\left(ik|\mathbf{x}| - \frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y})$$

The main idea is that for the Coulomb potential we have for large $\mathbf{y}$

$$V(\mathbf{y}) \approx 0$$

Hence we can restrict our attention to the $\mathbf{y}$ -ball such that

$$|\mathbf{y}| \ll |\mathbf{x}|$$

and let $V(\mathbf{y})$ censor everything else.

Consider the following expansions.

$$\begin{aligned} |\mathbf{x} - \mathbf{y}|^2 &= |\mathbf{x}|^2 - 2\mathbf{x} \cdot \mathbf{y} + |\mathbf{y}|^2 \\ \left(|\mathbf{x}| - \frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right)^2 &= |\mathbf{x}|^2 - 2\mathbf{x} \cdot \mathbf{y} + \left(\frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right)^2 \end{aligned}$$

Discard the $\mathbf{y}^2$ terms to obtain

$$|\mathbf{x} - \mathbf{y}|^2 = \left(|\mathbf{x}| - \frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right)^2$$

Hence for $\mathbf{x} \gg \mathbf{y}$

$$|\mathbf{x} - \mathbf{y}| \approx |\mathbf{x}| - \frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}$$

It follows that

$$\exp(ik|\mathbf{x} - \mathbf{y}|) \approx \exp\left(ik|\mathbf{x}| - \frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) \quad (1)$$

For the factor

$$\frac{1}{|\mathbf{x} - \mathbf{y}|}$$

we discard $\mathbf{y}$ altogether and use

$$\frac{1}{|\mathbf{x} - \mathbf{y}|} \approx \frac{1}{|\mathbf{x}|} \quad (2)$$

Hence by (1) and (2) we have

$$\frac{\exp(ik|\mathbf{x} - \mathbf{y}|)}{|\mathbf{x} - \mathbf{y}|} V(\mathbf{y}) \approx \frac{1}{|\mathbf{x}|} \exp\left(ik|\mathbf{x}| - \frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y})$$